Robust Deepfake Detection Under Real-World Perturbations

Anonymous Authors
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***Abstract*—This paper studies robust detection of manipulated face videos with a lightweight network. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise.**

***Index Terms*—deepfake detection, forensics, robustness**

I. Introduction

Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise.

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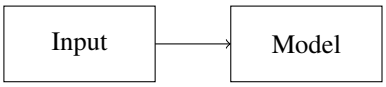


Fig. 1. Overview of the proposed pipeline.

TABLE I
Comparison with prior methods

Method	Acc	AUC
hline Baseline	81.2	0.85 Ours
93.7	0.96	
hline		

II. Method

Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. The loss is

$$L = \sum_i (y_i - \hat{y}_i)^2 \tag{1}$$

and the score is

$$s = \sigma(w^T x + b). \tag{2}$$

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III. Experiments

Deepfake detection has become an important problem in multimedia forensics because modern generators produce

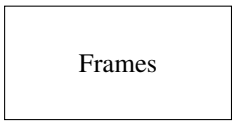


Fig. 2. Example frames.

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IV. Conclusion

Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise. Deepfake detection has become an important problem in multimedia forensics because modern generators produce faces that are hard to distinguish from real ones. In this work we study detectors under realistic perturbations such as compression, resizing and noise.

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References

[1] A. One, “Paper one,” *IEEE Trans. Inf. Forensics Security*, vol. 1, 2020.
[2] B. Two, “Paper two,” in *Proc. CVPR*, 2021.
[3] C. Three, “Paper three,” in *Proc. ICASSP*, 2022.
[4] D. Four, “Paper four,” *IEEE Access*, 2023.
[5] E. Five, “Paper five,” in *Proc. ICIP*, 2024.